

Jamhuriya RMS

System Documentation

Research Management System — Full Guide

Jamhuriya University of Science & Technology

Generated: 7/5/2026, 8:05:49 PM
Stack: MERN (MongoDB, Express, React, Node.js)
See also: docs/DATABASE_STRUCTURE.pdf

1. System Overview

Jamhuriya RMS is a web-based research management platform covering the full research lifecycle: proposal submission, ethics review (REC), project tracking, grants and budgets, publications, institutional repository, collaboration, and analytics.

Architecture: React SPA (Vite) !' Express REST API !' MongoDB. JWT authentication with HTTP-only refresh cookies. Role-based access control (RBAC) on both frontend routes and backend endpoints.

2. Technology Stack

Layer	Technology	Notes
Frontend	React 19 + Vite	Port 5173, proxies /api to backend
Backend	Express 5 + Node.js	Port 5000
Database	MongoDB + Mongoose	Database: rms
Auth	JWT + bcrypt	Access token + refresh cookie
Charts	Recharts	Dashboard analytics
PDF	PDFKit	Reports, ethics certificates, exports
Files	Multer	Proposal docs, repository uploads

3. User Roles

Role	Code	Responsibilities
Research Director	research_director	Institutional oversight, ethics approval, grants, user management, departments, analytics Faculty proposal pre-review, publication validation, faculty reports, audit trail
Faculty Coordinator	faculty_coordinator	
Finance Officer	finance_officer	
Ethics Committee	ethics_committee	Budgets, payment disbursement, finance reports, donor reports, closure finance approval JUREC ethics review and approval (/ethics)
Procurement Officer	procurement_officer	Purchase order review before director (/budgets queue)
Researcher	researcher	Proposals (budget, compliance docs), ethics forms, projects, publications, groups, repository uploads

4. Frontend Pages

Route	Page	Access
/dashboard	Dashboard & Analytics	All roles — role-specific widgets
/ethics	Ethics (REC)	Director, Coordinator, Researcher, Ethics Committee
/proposals	Proposals	Director, Coordinator, Researcher
/proposals/new	New Proposal + budget + compliance docs + Ethics	Researcher only
/proposals/:id/review	Director/Coordinator Review	Director, Coordinator
/projects	Projects	Director, Coordinator, Researcher
/projects/:id	Project Details — work plan, activities, comm log, closure	Director, Coordinator, Researcher
/publications	Publications & Outputs	Director, Coordinator, Researcher
/research-workflow	Research Workflow	Director, Coordinator, Researcher
/thesis	Thesis Groups (min 4 students)	Director, Coordinator, Researcher
/funding-calls	Funding Calls	Director, Coordinator, Finance, Researcher
/grants	Grants	Director, Coordinator, Finance, Researcher
/budgets	Finance & Budgets + Procurement PO queue	Director, Finance, Researcher, Procurement Officer
/finance-reports	Finance Reports	Director, Finance
/donor-reports	Donor Reports	Director, Finance
/audit-trail	Audit Trail	Director, Coordinator
/repository	Repository	Director, Coordinator, Researcher

/collaboration	Collaboration Hub	All authenticated roles
/groups	Research Groups	Director, Coordinator, Researcher
/messages	Messages	All authenticated roles
/notifications	Notifications	All authenticated roles
/pending-users	User Approval	Director only
/departments	Departments CRUD	Director only
/profile	Profile	All roles

5. REST API Endpoints

Base URL: <http://localhost:5000>. All routes except `/api/auth/register`, `/api/auth/login`, `/api/health`, and `/api/repository/oai` require authentication.

/api/auth

- POST `/register` — self-registration (pending approval)
- POST `/login` — JWT access + refresh cookie
- POST `/logout` — clear session
- POST `/refresh` — renew access token
- GET `/me` — current user profile

/api/users

- GET `/` — list users (director)
- GET `/pending` — pending registrations (director)
- POST `/:id/approve` | `/:id/reject` — user approval
- PUT `/me` — update own profile

/api/proposals

- GET `/` — list proposals (role-filtered)
- POST `/` — create proposal (researcher, multipart document)
- PUT `/:id` — update draft/revision (researcher)
- POST `/:id/submit` — submit proposal + linked ethics to director
- POST `/:id/review` — coordinator recommendation
- POST `/:id/director-decision` — approve/reject (creates project on approve)
- GET `/:id/ethics-application` — linked ethics form

/api/ethics

- GET `/` — list ethics applications
- POST `/` — create standalone ethics application
- PATCH `/:id` — save draft
- POST `/:id/submit` — submit to REC
- POST `/:id/director-decision` — approve/reject + issue certificate
- GET `/:id/certificate.pdf` — download ethics certificate

/api/projects

- GET `/` — list projects
- GET `/:id` — project details with PI
- PUT `/:id` — update project (director/researcher)
- POST `/:id/progress` — add progress report (researcher)

/api/publications

- GET / — list publications
- POST / — create publication (researcher)
- PUT /:id — update draft
- POST /:id/submit — submit for validation
- POST /:id/validate | /:id/reject — coordinator/director validation
- GET /citation?doi= — CrossRef DOI lookup

/api/grants

- GET / — list grants
- POST / — create grant request (researcher)
- PUT /:id — update grant
- POST /:id/submit — submit for review
- POST /:id/director-decision — approve with awarded amount

/api/budgets

- GET / — list budgets
- POST / — create budget (director/finance)
- POST /:id/items — add budget line item
- PATCH /:id/items/:itemId — approve/reject/pay item

/api/payments

- GET / — list payment requests
- POST / — create payment request
- POST /:id/director-decision — director approve/reject
- POST /:id/finance-pay — finance marks as paid

/api/procurement

- GET / — list purchase orders
- POST / — create PO
- POST /:id/director-decision — director approve/reject
- POST /:id/finance-pay — finance marks as paid

/api/repository

- GET / — list repository items
- POST / — upload file (PDF/CSV/Excel only)
- GET /export/csv | /export/pdf | /export/excel — catalog export
- GET /oai — OAI-PMH 2.0 (public, app-level route)
- GET /oai/export — OAI bulk export

/api/groups

- GET / — list research groups
- GET /stats — collaboration vs thesis counts
- POST / — create group
- POST /:id/join | /:id/leave — membership
- DELETE /:id — delete group (director)

/api/thesis-groups

- GET / — list thesis groups
- POST / — create thesis group
- PATCH /:id — update group
- POST /:id/meetings — log supervision meeting
- DELETE /:id — delete (director)

/api/conversations

- GET / — list my conversations
- GET /users — messageable users list
- POST / — start 1:1 conversation
- GET /group/:groupId — open group chat
- POST /:id/messages — send message (+ notification)

/api/notifications

- GET /me — my notifications
- GET /me/unread-count — badge count
- POST /:id/read — mark as read

/api/analytics

- GET /dashboard — role-based dashboard metrics
- GET /institutional — director institutional analytics
- GET /faculty-report — coordinator faculty report
- GET /faculty-report.pdf — faculty report PDF export
- GET /finance-report — finance summary
- GET /annual-report.pdf — director annual report PDF

/api/departments

- GET / — list departments
 - POST / — create (director)
 - PUT /:id — update (director)
 - DELETE /:id — delete (director)
-

6. Key Workflows

A. Proposal + Ethics (Researcher !' Director)

1. Researcher creates Proposal + Ethics form on one page (/proposals/new).
2. System auto-fills ethics from profile and proposal fields.
3. Researcher saves draft or submits both to Director in one action.
4. Director reviews ethics certificate + proposal on /proposals/:id/review.
5. On approval: Project is created automatically; notifications sent.

B. Publications (Researcher !' Coordinator/Director)

1. Researcher creates publication record (draft).
2. Submits for validation; appears in faculty workflow pipeline.
3. Coordinator or Director validates or rejects.
4. Validated publications count toward faculty analytics.

C. Finance (Researcher !' Director !' Finance)

1. Grant approved with awarded amount !' Budget created.
2. Researcher requests Payment or Purchase Order against budget.
3. Director approves request.
4. Finance Officer marks payment as paid / disbursed.

D. Collaboration & Messaging

1. Users create or join Research Groups (/groups).
2. Group chat opens via Conversation linked to groupId.
3. Direct 1:1 messaging via /messages with user picker.
4. Each message triggers a Notification for recipients.

E. Thesis Supervision

1. Coordinator/Director creates Thesis Group with students + supervisor.
 2. Optional link to ResearchGroup for chat.
 3. Supervisor logs meetings (date, location, agenda, notes).
 4. Status tracked: proposed !' in_progress !' submitted !' defended !' completed.
-

7. Database (MongoDB)

Database name: rms | URI: mongodb://localhost:27017/rms | Collections (15):

users, departments, proposals, ethicsapplications, projects, publications, grants, budgets, payments, purchaseorders, researchgroups, thesisgroups, conversations, notifications, repositoryitems

Full field-level documentation: docs/DATABASE_STRUCTURE.pdf (generate with `npm run docs:database-pdf`).

8. Initial User Bootstrap

Run `npm run seed` once to create institutional accounts (director, coordinators, finance officers, researchers).

Configure credentials via `SEED_*` variables in `backend/.env` or edit `src/scripts/seedData.js`.

The seed command does not create fictional proposals, grants, or publications.

9. Setup & Run

Prerequisites: Node.js 18+, MongoDB running locally.

Backend: `cd backend !' cp .env.example .env !' npm install !' npm run seed !' npm run dev`

Frontend: `cd frontend !' npm install !' npm run dev`

App URL: `http://localhost:5173` | API: `http://localhost:5000/api/health`

10. Export Features

- Repository catalog: PDF, Excel, CSV (/repository page buttons)
- Ethics certificate: PDF per approved application
- Faculty research report: PDF for coordinators
- Annual institutional report: PDF for director
- OAI-PMH 2.0: `/api/repository/oai` (institutional repository integration)

